Physical reachability of layer-wise training targets in block-MZI optical neural networks: supplemental document

BOYU ZHU, FENGHAI LIU, CONG XU, CHAORONG ZHOU, AND GUOZHU JIA*

Sichuan Normal University, Chengdu, China
**309739124@qq.com*

This document supports the main text. Equations, tables, figures and sections of the main text are cited as "main-text Eq. (1)", "main-text Table 3" and so on; items of this document carry the prefix S. Section S1 gives the modeling assumptions; Section S2 the certificate implementation, the implemented-encoding recomputation, the rounding-error-aware recheck and the regression baseline; Sections S3–S5 the block-size study, relaxation diagnostics and further target families; Sections S6–S8 the zeroth-order, simulated-chip and optical-power experiments; Section S9 the comparison with the published DGONN; and Section S10 the pre-specified tests, gate amendments and interpretation rules.

S1. Modeling assumptions and data protocol

Electronic stage clamp$(sI, 0, 1)$ with a scalar gain (trainable with exact gradients; fixed after calibration in SPSA; Eq. (S57) of Ref. [1] has none); fixed kept ports (equivalent under trainable U); three-module summed output layer (inferred from parameter counts); final-layer distillation by τ^2KL ($\tau = 2$) plus cross-entropy instead of the squared error of Ref. [1], because teacher logits are signed while intensities are not; feature normalization by the 99.5th percentile with clipping to $[0,1]$; exact gradients in the main experiments (local updates coincide with noiseless forward differences in the limit of small steps); nearest-neighbour heater crosstalk as a tunable systematic bias. Intensity encoding uses $\sqrt{x+\epsilon}-\sqrt{\epsilon}$ with $\epsilon = 10^{-12}$, which keeps zero inputs exactly dark (also in floating point) and the gradient bounded; certificates are evaluated in double precision from the single-precision cached features with the same formula. The rank-12 head teacher factorizes every dense head layer into two bias-free matrices of rank 12 ($17{,}016$ parameters) and shares the front end of the strong teacher. Static chip errors in zeroth-order training are applied to every perturbed measurement (unit-tested against the exact forward model).

Device-error model (Sec. S7): every phase shifter of every block (15 per block, including the three output phases that do not affect intensities) receives an independent static offset $\delta\phi \sim \mathcal{N}(0, \sigma_\phi^2)$, $\sigma_\phi = 0.05\pi$, added to the programmed phase after wrapping to $[0,2\pi)$ (and after quantization and heater crosstalk where these are enabled; they are not in the device-error study). Each of the two directional couplers of every MZI splits power as $0.5 \pm \delta$ with $\delta \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 0.03$, truncated at ± 0.45 (never reached at this σ). The static errors are drawn once per chip seed and layer and stay fixed for all measurements on that chip, including every perturbed measurement; measurement noise is drawn independently for every measurement. The implementation uses the Gaussian approximation

$$\tilde{I} = \max\{0, I + \sqrt{I/N_{ph}}Z\}, \qquad Z \sim \mathcal{N}(0,1), \qquad \text{(S1)}$$

with $N_{ph} = 10^4$ unless stated. It is not an exact Poisson photon-count simulation; the Gaussian approximation and non-negative clipping can be inaccurate when $N_{ph}I$ is small. When read noise is enabled, an independent Gaussian term with standard deviation equal to the configured fraction of the sample's peak measured intensity is added before the final non-negative clamp. Hidden gains in SPSA are calibrated on 2048 training samples to the configured output scale and then fixed; the output gain is updated electronically. Exact-

gradient methods optimize all gains. Device-adaptation runs initialize gains from the transferred network instead of recalibrating them.

Methods and codes. Besides E2E, LD-chain and LD-TF (main-text Table 2), the secondary experiments use LD-G (layer-wise distillation losses with global gradients) and LT-codes (layer-wise MSE to fixed random class codes [2,3]). The internally pre-specified amplitude-encoding matrix retains its original method codes: B1 = E2E, B2-G = LD-G, B2-CH = LD-chain, B2-TF = LD-TF, B5 = LT-codes, B4 = LD-chain for the first 30% of updates and cross-entropy on the output layer afterwards. P1 and P2 anneal the electronic-teacher contribution toward targets from an exponential-moving-average (EMA) copy of the ONN, averaged as unit phasors [4,5]: P1 maps the next deeper EMA output to each hidden layer, whereas P2 uses same-depth outputs. P3 adds 4-bit phase quantization, static phase errors ($\sigma_\phi = 0.05\pi$) and Gaussian-approximated shot noise ($N_{ph} = 10^4$) during P2 training.

Data protocol. MNIST [6] and Fashion-MNIST [7] use 55,000 training, 5000 validation and 10,000 test examples; CIFAR-10 [8] uses 45,000, 5000 and 10,000, respectively. The fixed validation split is sampled from the official training set with seed 0. FSDD [9] uses recording indices 0–4 for each speaker/digit as test (300 examples), and 5–49 as training (2700); speakers are shared across the split. FSDD has no separate validation subset and uses the frozen MNIST hyperparameters. Audio is converted to a 48-band by 48-frame log-mel image over a 0.9-second window. The separate "paper" protocol takes 10,000 examples from the original training set and splits them 7000/3000; no test sample is used for assignment search. Each teacher/cache is shared by its student seeds. Cache preprocessing uses training percentiles only.

Software. The recorded environment is listed in requirements-lock.txt; the reference machine uses an RTX 5060 Laptop GPU. REPRODUCE.md specifies the regeneration commands. The release manifest maps source code, raw JSON, generated assets and PDFs to hashes. Raw training results are unchanged by the certificate recomputations reported here.

S2. Certificate implementation and audit

S2.1 Linear and clipped readouts

Linear readout: feasible set $\{(M_b)_b, s\colon M_b \in s\mathcal{F}_k,\ 0 \le s \le S\}$; objective $f(M) = \sum_x \sum_b (a_b^\top M_b a_b - \tau_b)^2$ in the 10-dimensional coordinates of symmetric 4×4 matrices (general m: in matrix space, with gradient restarts [10]); FISTA [11] with step $1/(4\lambda_{\max}(Q))$, 400 iterations; golden-section search on $s \in [0, 8s_0]$, 40 iterations (the partial minimum $h(s)$ is convex); projection onto $s\mathcal{F}_k$ by an eigendecomposition followed by a capped-simplex projection of the eigenvalues; certificate LB = $\max\{0, f(M^\star) - g\}/(k \sum \| T\|^2)$ with Frank–Wolfe gap [12] $g = \langle \nabla f, M^\star \rangle - \min(0, S \sum_b \lambda_{1..k}(\nabla f_b))$, $S = 4s^\star$; the cap is removed when a certified lower bound on $h(S)$ exceeds $f(M^\star)$, which holds for every layer (Table S2). The block-size study (Sec. S3) uses the matrix-space implementation. Lower bounds are in double precision, normalized by the target energy of the 5000 samples and rounded down; fitted residuals and primal values use ordinary rounding. Unit tests check that certificates vanish for targets generated by actual (clipped) block layers and stay below directly fitted residuals.

Same-family check. Table S1 gives, for each setting and certificate, the smallest ratio over the five hidden layers of the directly fitted residual in the certificate's own readout family to the certificate (fits: 1500 steps, two starts, teacher-forced inputs, same 5000 samples; ratios rounded down). Scalar-gain linear fits are compared with $\mathrm{LB}^{\mathrm{lin}}$ and with $\mathrm{LB}^{\mathrm{free}}$ (a scalar gain is a special case of non-negative per-port gains), per-port non-negative linear fits without offset with $\mathrm{LB}^{\mathrm{free}}$ of the same grouping, clipped fits with $\mathrm{LB}^{\mathrm{clamp}}$, and the smaller of the linear and clipped fits with the dark-block term without offset. All *120* comparisons give ratios above one; the smallest, *1.01*, belongs to the scalar linear readout. The intensity-encoding rows use the certificates recomputed with the implemented encoding (Sec. S2.4).

Table S1. Certificates versus directly fitted residuals of the same readout family: smallest ratio fitted/certified over the hidden layers (rounded down; –: no fit of that family; dark: dark-block term without offset).

Setting	$G^{\mathrm{lin}}/\mathrm{LB}^{\mathrm{lin}}$	$G^{\mathrm{lin}}/\mathrm{LB}^{\mathrm{free}}$	$G^{\mathrm{lin}}_{\mathrm{pp}}/\mathrm{LB}^{\mathrm{free}}$	$G_{\mathrm{TF}}/\mathrm{LB}^{\mathrm{clamp}}$	fit/dark
intensity, native	1.01	4.26	3.92	1.81	4.47
intensity, optimized	1.13	1.32	1.17	1.37	52.39
amplitude, native	1.01	3.46	3.25	1.76	5.29
amplitude, optimized	1.09	1.24	1.09	1.39	70.76
amplitude, output only	–	–	–	1.44	36.33
amplitude, grouping only	–	–	–	1.16	3.01

Table S2. Linear-readout certificates per layer (amplitude encoding, native assignment).

Layer	$s^\star$	S	$f(M^\star)$	FW gap	$\mathrm{LB}^{\mathrm{lin}}$	LB on $h(S)$	certified
MNIST L1	1.578	6.310	0.3722	3.1e-04	0.3719	1.090	yes
MNIST L2	3.647	14.587	0.5393	7.5e-03	0.5317	1.018	yes
Fashion-MNIST L1	1.581	6.325	0.3496	2.9e-04	0.3493	0.922	yes
Fashion-MNIST L2	2.479	9.914	0.2818	8.0e-04	0.2809	0.723	yes
Fashion-MNIST L3	2.987	11.950	0.3202	3.4e-03	0.3167	0.824	yes

Clamp-aware certificates. Tables S3 and S4 (amplitude and intensity encoding, native assignment) give the interval with the smallest certificate, its position relative to the optimal linear gain $s^\star$, the primal value on it, the certificate restricted to gains $s \le 10s^\star$ (Sec. S2.2) and the numbers of intervals and refinements. The maximum number of interval refinements differs between studies: 25 for amplitude encoding with the native assignment, 15 for amplitude encoding with the optimized assignment and for intensity encoding with the native assignment, and 10 for intensity encoding with the optimized assignment, for the amplitude-encoding output-only and grouping-only variants and for the ONN-teacher targets. Each value is a valid bound for its stated setting; values obtained with different limits, or on intervals with large Frank–Wolfe gaps (Sec. S2.4), are not directly comparable to the third decimal.

Table S3. Clamp-aware certificates (amplitude encoding, native assignment; at most 25 refinements). $s^\star$: optimal gain of the linear readout; $s_1/s^\star$: position of the interval with the smallest certificate; $\mathrm{LB}^{\mathrm{clamp}}_{s\le 10s^\star}$: certificate for gains up to $10s^\star$.

Layer	$\mathrm{LB}^{\mathrm{lin}}$	Primal	$\mathrm{LB}^{\mathrm{clamp}}$	$\mathrm{LB}^{\mathrm{clamp}}_{s\le 10s^\star}$
MNIST L1	0.371	0.336	0.335	0.335
MNIST L2	0.531	0.387	0.270	0.349
Fashion-MNIST L1	0.349	0.394	0.283	0.288
Fashion-MNIST L2	0.280	0.236	0.187	0.191
Fashion-MNIST L3	0.316	0.172	0.172	0.200

Layer	$s^\star$	Minimum interval $[s_1, s_2]$	$s_1/s^\star$	Intervals / splits
MNIST L1	1.58	[*12.6,17.8*]	8	39 / 25
MNIST L2	3.65	[*1.26e + 04,1.31e + 04*]	3.44e+03	27 / 13

Layer	$s^\star$	Minimum interval $[s_1, s_2]$	$s_1/s^\star$	Intervals / splits
Fashion-MNIST L1	1.58	[*1.3e + 04*,*1.54e + 04*]	8.19e+03	39 / 25
Fashion-MNIST L2	2.48	[*6.04e + 03*,*6.3e + 03*]	2.44e+03	32 / 18
Fashion-MNIST L3	2.99	[*4.89e + 04*, ∞]	1.64e+04	19 / 5

Table S4. As Table S3, intensity encoding as implemented (native assignment; at most 15 refinements).

Layer	LB^{lin}	Primal	LB^{clamp}	$LB^{clamp}_{s\le 10s^\star}$
MNIST L1	0.299	0.276	0.276	0.276
MNIST L2	0.432	0.540	0.228	0.280
Fashion-MNIST L1	0.268	0.236	0.235	0.235
Fashion-MNIST L2	0.214	0.224	0.160	0.165
Fashion-MNIST L3	0.235	0.174	0.152	0.165

Layer	$s^\star$	Minimum interval $[s_1, s_2]$	$s_1/s^\star$	Intervals / splits
MNIST L1	1.24	[*0.619*,*0.876*]	0.5	29 / 15
MNIST L2	2.18	[*4.46e + 03*,*8.92e + 03*]	2.05e+03	29 / 15
Fashion-MNIST L1	1.24	[*0.621*,*0.879*]	0.5	29 / 15
Fashion-MNIST L2	1.52	[*389*,*1.56e + 03*]	256	29 / 15
Fashion-MNIST L3	1.85	[*5.36e + 03*,*5.85e + 03*]	2.9e+03	29 / 15

Table S5. Assignment-free certificates (amplitude encoding): primal value $\sum_j \min_b$ of the per-unit fits and certified bound.

	native input grouping		optimized input grouping	
Layer	primal $\sum_j \min_b$	certified LB^{free}	primal	certified
MNIST L1	0.1516	0.151	0.1123	0.112
MNIST L2	0.1002	0.099	0.0659	0.065
Fashion-MNIST L1	0.1642	0.164	0.1319	0.131
Fashion-MNIST L2	0.1022	0.102	0.0751	0.074
Fashion-MNIST L3	0.0940	0.093	0.0662	0.066

S2.2 Convex envelope for the clipped readout

On an interval $[s_1, s_2]$, $D(Y)$ of main-text Proposition 2 consists of an undershoot part $(\tau - s_2 Y)^2$ for $Y < \tau/s_2$ (convex), a zero part, and, if $\tau < 1$, an overshoot part $(\min(s_1 Y, 1) - \tau)^2$ for $Y > \tau/s_1$, which rises and then becomes flat. If $s_1 \parallel a\parallel^2 > 1$ the flat part is replaced by the tangent from $(\parallel a\parallel^2, (1-\tau)^2)$ to the parabola $(s_1 Y - \tau)^2$, touching at $s_1 Y_0 - \tau = w$ with $w = (s_1 \parallel a\parallel^2 - \tau) - \sqrt{(s_1 \parallel a\parallel^2 - \tau)^2 - (1-\tau)^2}$. The envelope is C^1, convex and below D (unit-tested). On the tail interval $s_2 = \infty$ the upper end of the interval for Z is $\min(s_2 Y, k) = k$ for every $Y > 0$ (and 0 at $Y = 0$), so $D(Y) = \mathrm{dist}(\tau, [\min(s_1 Y, 1), k])^2$ for $Y > 0$, while at $Y = 0$ the kept ports are dark and $Z = 0$. The implementation keeps the undershoot term τ^2 only for

samples whose block input vanishes ($\| a\|^2 = 0$) and sets it to zero otherwise, which can only lower the bound. The initial cover uses intervals with ratio 2 on $[s^\star/64, 64s^\star]$ plus $[0, s^\star/64]$ and $[64s^\star, \infty)$; the interval with the smallest certificate is split geometrically (tail intervals as $[s_1, 4s_1] \cup [4s_1, \infty)$) up to the refinement limit of Sec. S2.1; intervals far from $s^\star$ are solved with several step scales and the largest certificate is kept. With the native assignment, the smallest certificate lies at very large gains in *4* of 5 layers with amplitude encoding ($s_1/s^\star$ = *2435–16384*, where the clipped readout is nearly binary) and in *3* of 5 with intensity encoding (*256–2896*); with the optimized assignments its interval starts at $s_1 \le 1.6\, s^\star$. Restricting the gain to $s \le 10 s^\star$ raises the certificate by at most *0.079* (largest in MNIST L2, *0.270* → *0.349*) and *0.053* (MNIST L2, *0.228* → *0.280*); the main text quotes the unrestricted values. For targets from a clipped ONN, a positive linear bound can coexist with a near-zero clipped residual (main-text Table 4), which is why the two readout families are treated separately. With a free per-port offset, the dark-block bound becomes the target variance on that block's dark samples; these contributions depend on the output assignment, and their minima over assignments are small (Sec. S2.3).

S2.3 Assignment-free bound and assignment optimization

For one block and target, let $\mathcal{A}(M)_x = a_x^\top M a_x$. The PSD least-squares problem has residual-space dual

$$\min_{M \succcurlyeq 0} \| \mathcal{A}(M) - t\|^2 \;\ge\; -\| y\|^2 - 2t^\top y, \qquad \mathcal{A}^*(y) = \sum_x y_x\, a_x a_x^\top \succcurlyeq 0. \tag{S2}$$

To see this, expand $\| \mathcal{A}(M) - t - y\|^2 \ge 0$ and use $\langle \mathcal{A}^*(y), M\rangle \ge 0$. We construct three residual candidates: the residual of the 400-step FISTA PSD iterate, the unrestricted symmetric least-squares residual (computed by direct SVD of the quadratic-feature matrix), and the residual at a quadratic Lagrangian minimizer defined below. For each candidate r, set $C = \sum_x a_x\, a_x^\top$ and $B = \mathcal{A}^*(r)$. On nonzero input coordinates, $C = LL^\top$ and

$$\alpha \ge \max\{0, -\lambda_{\min}(L^{-1} B L^{-\top})\}, \qquad y = r + \alpha \mathbf{1} \tag{S3}$$

ensures $\mathcal{A}^*(y) = B + \alpha C \succcurlyeq 0$. Scaling y by $\beta = \max\{0, -t^\top y/\| y\|^2\}$ maximizes the dual along its non-negative ray (use zero for a zero vector). The implementation removes only identically zero input coordinates, adds a positive $10^{-10}\max(1, \| L^{-1} B L^{-\top}\|_2)$ margin, and recomputes the adjoint from explicit residuals to check feasibility. An entirely dark block returns $\| t\|^2$; unresolved singular or ill-conditioned input Gram matrices return zero. For each block and unit we retain the maximum non-negative checked dual value, including a single-residual implementation as a floor, before taking $\sum_j \min_b$ and normalizing by target energy. Thus no individual candidate can weaken the aggregate bound.

For the third candidate, write $\mathcal{A}(M) = \Phi m$, $Q = \Phi^\top \Phi$, $c = \Phi^\top t$, and $\langle Z, M\rangle = z^\top m$. We take Z as the PSD part of the gradient at the primal iterate and set $m_{\mathrm{L}} = Q^\dagger (c + z/2)$. If the exact stationarity condition $Q m_{\mathrm{L}} = c + z/2$ holds, its residual satisfies $\mathcal{A}^*(\Phi m_{\mathrm{L}} - t) = Z/2 \succcurlyeq 0$ and the unshifted residual dual equals the quadratic Lagrangian dual. The candidate matrix need not itself be PSD. In numerical use, neither a small range residual nor the least-squares normal equations establish acceptance: all three candidates pass through the explicit shift and adjoint check above.

All *20* encoding/layer/grouping configurations give positive bounds, with minimum *0.034* and *0* fallback blocks (Table S6). Each is at least as large as the single-residual dual bound and the signed least-squares value, and at least *99.6*% of its corresponding relaxed primal objective. Compared with an earlier pseudoinverse estimate, the largest absolute change is at most *0.00010552* and the largest relative change at most *0.16153*%. That estimate accepted range residuals below 10^{-8} on a potentially rank-deficient quadratic-feature Gram matrix; this tolerance alone does not establish exact dual feasibility on an unbounded cone, and the estimate is not used as a certificate.

Assignment search uses unconstrained quadratic costs for eight rounds of 3000 input-swap proposals, alternating with Hungarian output assignments (k slots per block), then a final Hungarian assignment on PSD primal costs. "Output only" and "grouping only" freeze the other variable. Routes $r_i = u_i^{-1}[g_i]$ are applied identically to chained and teacher-forced inputs. The 7000/3000 protocol repeats search on its own teacher's training cache. Table S7 compares the optimized linear per-port fit with the assignment-free bound for the same grouping, for both encodings.

Table S6. Assignment-free numerical audit of all 20 configurations. Enc.: amp, amplitude encoding; sqrt, intensity encoding as implemented. Pinv. est.: pseudoinverse estimate (not a certificate); Single dual: single-residual dual; Signed LS: unrestricted symmetric least squares, Q^{free}; Final cert.: maximum of independently checked residual candidates, used throughout; Primal: relaxed fitted objective. Lower values are rounded down.

Layer	Enc.	Group	Pinv. est.	Single dual	Signed LS	Final cert.	Primal
MNIST L1	amp	Native	0.151	0.117	0.145	0.151	0.152
MNIST L1	amp	Opt.	0.112	0.065	0.107	0.112	0.112
MNIST L2	amp	Native	0.099	0.036	0.091	0.099	0.100
MNIST L2	amp	Opt.	0.065	0.025	0.057	0.065	0.066
F-MNIST L1	amp	Native	0.164	0.100	0.160	0.164	0.164
F-MNIST L1	amp	Opt.	0.131	0.089	0.127	0.131	0.132
F-MNIST L2	amp	Native	0.101	0.040	0.095	0.102	0.102
F-MNIST L2	amp	Opt.	0.074	0.036	0.068	0.074	0.075
F-MNIST L3	amp	Native	0.093	0.093	0.085	0.093	0.094
F-MNIST L3	amp	Opt.	0.066	0.065	0.057	0.066	0.066
MNIST L1	sqrt	Native	0.097	0.097	0.093	0.097	0.098
MNIST L1	sqrt	Opt.	0.078	0.078	0.073	0.078	0.078
MNIST L2	sqrt	Native	0.065	0.046	0.058	0.065	0.066
MNIST L2	sqrt	Opt.	0.034	0.022	0.030	0.034	0.035
F-MNIST L1	sqrt	Native	0.110	0.092	0.107	0.110	0.111
F-MNIST L1	sqrt	Opt.	0.087	0.063	0.084	0.087	0.088
F-MNIST L2	sqrt	Native	0.068	0.060	0.062	0.068	0.068
F-MNIST L2	sqrt	Opt.	0.048	0.045	0.042	0.048	0.049
F-MNIST L3	sqrt	Native	0.062	0.061	0.052	0.062	0.062
F-MNIST L3	sqrt	Opt.	0.045	0.045	0.035	0.045	0.046

Table S7. Assignment heuristic versus the assignment-free certificate (optimized input grouping, linear readout with per-port non-negative gains): LB^{free} (rounded down), fitted residual $G_{\text{pp}}^{\text{lin}}$ of the optimized assignment (upper estimate) and their ratio.

	$a = \sqrt{x}$, optimized			$a = x$, optimized		
Layer	LB^{free}	$G_{\text{pp}}^{\text{lin}}$ fit	ratio	LB^{free}	$G_{\text{pp}}^{\text{lin}}$ fit	ratio
MNIST L1	0.078	0.092	1.17	0.112	0.132	1.18
MNIST L2	0.034	0.041	1.19	0.065	0.072	1.10

Fashion-MNIST L1	0.087	0.105	1.19	0.131	0.151	1.15
Fashion-MNIST L2	0.048	0.060	1.23	0.074	0.088	1.17
Fashion-MNIST L3	0.045	0.054	1.19	0.066	0.081	1.23

S2.4 Implemented versus ideal intensity encoding

The networks and fits use $\phi(x) = \sqrt{x+\epsilon} - \sqrt{\epsilon}$, whereas earlier certificate runs used the ideal $\sqrt{x}$. All intensity-encoding certificates were recomputed with the implemented ϕ, through the same code path as the fitted residuals (cached features, the trained student's own routing, teacher-forced inputs), with the same solver settings and refinement limits. Before the recomputation, the ideal encoding was passed through the new code path and reproduced the archived $\mathrm{LB}^{\mathrm{lin}}$ and $\mathrm{LB}^{\mathrm{free}}$ values exactly. Table S8 compares the two encodings. All implemented-encoding certificates are positive. For $\mathrm{LB}^{\mathrm{free}}$ the largest change is 4.1×10^{-8} absolute and 4.4×10^{-7} relative; for $\mathrm{LB}^{\mathrm{lin}}$ it is 5.0×10^{-6}. The amplitudes of the two encodings differ by at most $\sqrt{\epsilon} = 10^{-6}$, so changes of this size are expected. The clipped certificates fall into two groups. Where the smallest certificate lies near the linear optimum $s^\star$ (all optimized assignments and two native ones), the change is below 5.0×10^{-6}, as for all other certificates. For the three native-assignment layers whose smallest certificate lies at gains above $256\,s^\star$ (MNIST L2, F-MNIST L2 and F-MNIST L3), the certificates change by *0.0015–0.0085*, most for MNIST L2 (*0.236* → *0.228*). On these intervals the clipped readout is nearly binary and the Frank–Wolfe gap between primal value and certificate is *0.02–0.40* in normalized units, i.e. the inner solver is far from converged. Because each child interval is warm-started from its parent, a perturbation of the inputs at the 10^{-6} level changes which intervals are refined, hence the warm starts and the certificates obtained within the fixed refinement limit. Both sets of values are valid lower bounds for their encodings; the differences reflect incomplete convergence on these intervals, not a property of the encoding. The main text reports the implemented-encoding values.

Table S8. Certificates with the ideal encoding $\sqrt{x}$ and with the implemented encoding $\sqrt{x+\epsilon} - \sqrt{\epsilon}$ ($\epsilon = 10^{-12}$), unrounded. $\mathrm{LB}^{\mathrm{lin}}$ **and** $\mathrm{LB}^{\mathrm{clamp}}$ **use the stated assignment;** $\mathrm{LB}^{\mathrm{free}}$ **the stated grouping.**

		$\mathrm{LB}^{\mathrm{lin}}$		$\mathrm{LB}^{\mathrm{clamp}}$		$\mathrm{LB}^{\mathrm{free}}$	
Layer	Assignment/grouping	$\sqrt{x}$	implemented	$\sqrt{x}$	implemented	$\sqrt{x}$	implemented
MNIST L1	Native	0.299716	0.299716	0.276255	0.276255	0.097575	0.097575
MNIST L1	Optimized	0.058650	0.058650	0.044741	0.044741	0.078217	0.078217
MNIST L2	Native	0.432154	0.432155	0.236492	0.228030	0.065748	0.065748
MNIST L2	Optimized	0.042119	0.042119	0.034323	0.034323	0.034673	0.034673
F-MNIST L1	Native	0.268850	0.268845	0.235758	0.235759	0.110724	0.110724
F-MNIST L1	Optimized	0.064241	0.064241	0.047547	0.047547	0.087863	0.087863
F-MNIST L2	Native	0.214145	0.214145	0.159018	0.160558	0.068068	0.068068
F-MNIST L2	Optimized	0.030985	0.030985	0.021839	0.021839	0.048720	0.048720
F-MNIST L3	Native	0.235119	0.235119	0.154106	0.152371	0.062075	0.062075
F-MNIST L3	Optimized	0.026522	0.026522	0.018487	0.018487	0.045741	0.045741

S2.5 Rounding-error-aware recheck of the assignment-free certificates

The acceptance test of Sec. S2.3 uses a floating-point eigensolver with a positive margin. To remove the dependence on eigensolver accuracy, every dual vector that determines a

block/unit maximum in all *30* configurations (20 of the main audit, of which the 10 intensity-encoding configurations use the implemented encoding, plus the 10 archived ideal-encoding configurations) was rechecked as follows. The floating-point dual vector $\tilde{y}$ is recomputed from the stored candidate, shift and scale and is treated as an exact rational vector, so its dual value $-\|\tilde{y}\|^2 - 2t^\top\tilde{y}$ is a valid bound whenever $\mathcal{A}^*(\tilde{y}) \succcurlyeq 0$. The adjoint $\hat{S} = \mathrm{fl}(\sum_x \tilde{y}_x\, a_x a_x^\top)$ satisfies $|\hat{S} - \mathcal{A}^*(\tilde{y})| \le \gamma_{n+2} \sum_x |\tilde{y}_x||a_x||a_x|^\top$ elementwise, with $\gamma_k = ku/(1-ku)$, $u = 2^{-53}$ and any summation order [13]. With any nonsingular lower-triangular matrix K (the computed inverse Cholesky factor of C), $\widehat{W} = \mathrm{fl}(K\hat{S}K^\top)$ differs from $K\mathcal{A}^*(\tilde{y})K^\top$ elementwise by at most $|K|E_S|K|^\top + \gamma_{2m+2}|K||\hat{S}||K|^\top$, and the Frobenius norm δ of this bound limits the spectral error. Positive semidefiniteness of $\frac{1}{2}(\widehat{W} + \widehat{W}^\top) - \delta I$ is then decided exactly in rational arithmetic, which implies $\mathcal{A}^*(\tilde{y}) \succcurlyeq 0$ by congruence. The dual value is reduced by its own rounding-error bound, and the aggregate by bounds on the summation and division. Error bounds are inflated by 1%. Of *730,880* tested candidates none failed; the smallest ratio of the computed minimum eigenvalue to δ was *6*. Block/unit pairs whose maximum came from the single-residual floor (*20,132* pairs) were assigned the largest rechecked candidate instead, which lowers no pair by more than 3.0×10^{-9}. The rechecked aggregate bounds are at most 1.5×10^{-10} (relative) below the reported values. This check assumes IEEE-754 double-precision arithmetic with rounding to nearest; it is not an interval-arithmetic verification of the primal solver, which is not needed for the validity of a dual bound.

S2.6 Ordinary quadratic regression

For main-text Table 4, each teacher unit is regressed on the ten quadratic features $a_i a_j$ of the four inputs of a block by unconstrained least squares without offset, computed from the singular value decomposition of the feature matrix (relative cutoff 10^{-12}). Q uses the block of the given assignment; Q^{free} uses each unit's best block, as in main-text Eq. (4). Both are normalized by the target energy and lower-bound linear readouts with arbitrary per-port gains and no offset. They do not bound clipped readouts. Table S9 lists all values.

Table S9. Ordinary quadratic regression residuals (intensity encoding as implemented, rounded down): given assignment (Q) and best block for every unit (Q^{free}, "all").

	Electronic, native		Electronic, optimized		ONN teacher	
Layer	given	all	given	all	given	all
MNIST L1	0.445	0.093	0.076	0.073	0.025	0.025
MNIST L2	0.390	0.058	0.032	0.030	0.025	0.025
F-MNIST L1	0.419	0.107	0.089	0.084	0.011	0.011
F-MNIST L2	0.328	0.062	0.046	0.042	0.014	0.014
F-MNIST L3	0.372	0.052	0.041	0.035	0.023	0.021

S3. Block size

Larger blocks reduce the fitted residual (main-text Fig. 6, amplitude encoding): it reaches *0.036–0.139* at $m = 16$ and *0.011–0.028* at $m = 64$. Vanishing certificates at large m reflect both a weaker constraint and incomplete solver convergence. A positive value at $m = 16$ whose gain cap could not be removed is counted as zero. Its certificates at $m = 4$ exceed those of Table S2 (e.g. MNIST L2) because the matrix-space implementation converges to a much smaller duality gap; both lie below the primal objective (Table S11). These zero certificates do not establish reachability.

Table S10. Certified linear bound (rounded down) and fitted gap (clamp, upper estimate) versus block size m (amplitude encoding). †: a positive value whose gain cap could not be removed, counted as zero (Table S11).

	certified LB^{lin}					fitted G_{TF} (clamp, upper estimate)				
Layer	*m = 4*	*m = 8*	*m = 16*	*m = 32*	*m = 64*	*m = 4*	*m = 8*	*m = 16*	*m = 32*	*m = 64*
MNIST L1	0.372	0.131	0.029	0.003	0.000	0.592	0.323	0.128	0.049	0.024
MNIST L2	0.539	0.114	0.000	0.000	0.000	0.540	0.209	0.051	0.018	0.011
Fashion-MNIST L1	0.349	0.122	0.030	0.004	0.000	0.564	0.318	0.139	0.057	0.028
Fashion-MNIST L2	0.281	0.069	0.007	0.000	0.000	0.459	0.192	0.056	0.025	0.015
Fashion-MNIST L3	0.320	0.066	0.000 †	0.000	0.000	0.515	0.169	0.036	0.018	0.014

Table S11. Block-size certificates: certified bound plus Frank–Wolfe gap (equal to the primal objective where the bound is positive and an upper bound on it where the bound is zero), the gap, and the certified bound (same normalization). †: positive only for gains below the cap *S*, because the cap could not be removed; not a certificate for all gains and counted as zero.

Layer	quantity	*m = 4*	*m = 8*	*m = 16*	*m = 32*	*m = 64*
MNIST L1	LB + gap	0.3722	0.1312	0.0300	0.0061	0.0031
	FW gap	3.7e-05	1.6e-05	2.1e-04	2.5e-03	3.1e-03
	LB	0.372	0.131	0.029	0.003	0.000
MNIST L2	LB + gap	0.5392	0.1152	0.0183	0.0083	0.0088
	FW gap	9.1e-05	9.4e-04	1.8e-02	8.3e-03	8.8e-03
	LB	0.539	0.114	0.000	0.000	0.000
Fashion-MNIST L1	LB + gap	0.3496	0.1227	0.0310	0.0078	0.0032
	FW gap	5.1e-07	2.5e-06	2.6e-04	3.0e-03	3.2e-03
	LB	0.349	0.122	0.030	0.004	0.000
Fashion-MNIST L2	LB + gap	0.2818	0.0696	0.0127	0.0064	0.0058
	FW gap	1.3e-06	9.2e-05	5.6e-03	6.4e-03	5.8e-03
	LB	0.281	0.069	0.007	0.000	0.000
Fashion-MNIST L3	LB + gap	0.3202	0.0676	0.0093	0.0063	0.0070
	FW gap	1.7e-07	6.9e-04	5.2e-03	6.3e-03	7.0e-03
	LB	0.320	0.066	0.004 †	0.000	0.000

S4. Relaxation hierarchy and readout diagnostics

Removing a physical constraint enlarges the model family. For a fixed assignment and linear readout, passive lossy blocks are relaxed by positive-semidefinite quadratic forms; allowing signed forms or offsets enlarges the family further. The differences between their optima quantify changes in the relaxation, rather than a decomposition of the physical residual. Diagnostic lossy-block values (positive-semidefinite cone, *0.25–0.51*), per-port affine readout (*0.13–0.36*; fitted *0.30–0.41*) and teacher feature variants (*0.10–0.61*) are summarized in Tables S12 and S13. The planned lossy relaxation $\{0 \preccurlyeq A \preccurlyeq I, \mathrm{tr}A \leq k\}$ with a shared gain could not be certified (loss and gain trade off freely) and was replaced by the positive-semidefinite cone.

The PSD lossy/affine columns in this section use an approximate pseudoinverse-range acceptance (Sec. S2.3), with offsets where applicable. They are numerical dual estimates, not additional validated certificates, and are not used to establish the main unreachability result. The signed columns are unconstrained least-squares diagnostics. The mathematical nesting of these model families remains valid, but differences between these numerical columns are not certified decompositions of the physical gap. The data are preserved so that this qualification is visible.

Table S14 lists the relaxation hierarchy and fitted per-port residuals (amplitude encoding, native assignment). Without offsets, removing energy conservation (positive-semidefinite forms with a shared gain, lossy blocks) changes the dual estimate by at most *0.024* (*0.257–0.514* versus *0.280–0.531*). With per-port gains and offsets the dual estimates are *0.138–0.365* for passive blocks and $0.114-0.293$ for signed gains; the dark-block term is $0.031-0.111$ without offsets and *0.010–0.047* with offsets. The fitted columns are upper estimates: the best function of the four block inputs that our regressors find (held-out data) leaves *0.15–0.23* with the native and *0.020–0.069* with the optimized assignment, a signed quadratic form *0.25–0.34*, the physical layer $0.46-0.59$. Regressor residuals include fitting error, so they are not the irreducible error of block locality; we use them only to note that block-local functions can fit the targets considerably better than the physical layer, especially after the assignment is optimized.

Table S12. Readout diagnostics (amplitude encoding; see the numerical qualification above): scalar gain with lossless and lossy blocks, per-port affine readout with non-negative or signed gains, and directly fitted residuals.

Data	Layer	relaxation values				fitted $G_{\mathrm{TF}}^{\mathrm{lin}}$	
		scalar, lossless	scalar, lossy	per-port $s_p \geq 0$, c_p	per-port signed	scalar	per-port
MNIST	1	0.371	0.351	0.216	0.184	0.596	0.406
	2	0.531	0.514	0.365	0.293	0.540	0.367
Fashion-MNIST	1	0.349	0.329	0.188	0.167	0.569	0.368
	2	0.280	0.257	0.151	0.129	0.464	0.302
	3	0.316	0.299	0.138	0.114	0.519	0.326

Table S13. Certified bound and fitted gap on Fashion-MNIST for different teacher feature distributions (amplitude encoding).

	layer 1		layer 2		layer 3	
Teacher features	LB	G_{TF}	LB	G_{TF}	LB	G_{TF}
baseline (per-layer scalar)	0.349	0.565	0.280	0.459	0.316	0.515
(i) per-unit normalization	0.348	0.561	0.287	0.470	0.307	0.504
(ii) dense front end (no ReLU)	0.235	0.433	0.354	0.576	0.333	0.504
(iii) sigmoid teacher	0.616	0.711	0.152	0.247	0.103	0.183

Table S14. Relaxation hierarchy (amplitude encoding, native assignment, teacher-forced inputs). Readout diagnostics, rounded down, of nested relaxations within each readout family (block sums aggregated by Cauchy–Schwarz; dark-block terms exact per port), and fitted per-port residuals (upper estimates; "local": best of a neural and a quadratic regressor on held-out data).

Layer	Dark	PSD	Fantope	Clamp	Dark+offset	Signed+offset	PSD+offset
MNIST L1	0.111	0.351	0.371	0.335	0.047	0.184	0.216
MNIST L2	0.073	0.514	0.531	0.270	0.028	0.293	0.365
Fashion-MNIST L1	0.071	0.329	0.349	0.283	0.026	0.167	0.188
Fashion-MNIST L2	0.031	0.257	0.280	0.187	0.010	0.129	0.151
Fashion-MNIST L3	0.069	0.299	0.316	0.172	0.021	0.114	0.138

Layer	Local (native)	Local (optimized)	Signed quadratic	Physical clamp
MNIST L1	0.225	0.052	0.341	0.593
MNIST L2	0.158	0.021	0.294	0.538
Fashion-MNIST L1	0.233	0.069	0.324	0.565
Fashion-MNIST L2	0.152	0.027	0.252	0.459
Fashion-MNIST L3	0.145	0.020	0.267	0.515

S5. Extended target families and amplitude encoding

Tables S15–S17 split the full target-family matrix into readable parts. Every accuracy cell reports its own number of seeds; in particular, the rank-12 teacher has two LD-chain seeds and three E2E seeds. Dashes indicate a condition not run, not zero performance. The Fantope and clamp columns in Table S15 cover the stated assignment. Random class codes match or exceed electronic targets with native intensity assignment, but electronic targets improve LD-chain after assignment optimization.

Table S15. Complete target-family residuals. Encoding: int., intensity (as implemented); amp., amplitude. Bounds are means across hidden layers, rounded down; G is the mean fitted clipped residual.

Data	Enc.	Targets	$\overline{\mathrm{LB}}^{\mathrm{lin}}$	$\overline{\mathrm{LB}}^{\mathrm{clamp}}$	$\overline{G}_{\mathrm{TF}}$
MNIST	int.	Native	0.365	0.252	0.470
MNIST	int.	Optimized	0.050	0.039	0.078
MNIST	int.	ONN teacher	0.018	0.000	*< 0.001*
MNIST	amp.	Native	0.451	0.302	0.565
MNIST	amp.	Output optimized	0.123	0.083	0.157
MNIST	amp.	Grouping optimized	0.440	0.386	0.557
MNIST	amp.	Both optimized	0.086	0.057	0.112
MNIST	amp.	ONN teacher	0.059	0.000	*< 0.001*
F-MNIST	int.	Native	0.239	0.182	0.424
F-MNIST	int.	Optimized	0.040	0.029	0.084
F-MNIST	int.	ONN teacher	0.007	0.000	*< 0.001*
F-MNIST	amp.	Native	0.315	0.214	0.513

Data	Enc.	Targets	$\overline{\mathrm{LB}}^{\mathrm{lin}}$	$\overline{\mathrm{LB}}^{\mathrm{clamp}}$	$\overline{G}_{\mathrm{TF}}$
F-MNIST	amp.	Output optimized	0.098	0.054	0.153
F-MNIST	amp.	Grouping optimized	0.337	0.274	0.541
F-MNIST	amp.	Both optimized	0.072	0.039	0.115
F-MNIST	amp.	ONN teacher	0.038	0.000	*< 0.001*
F-MNIST	amp.	Sigmoid teacher	0.290	–	0.380
F-MNIST	amp.	Dense front end	0.307	–	0.505
F-MNIST	amp.	Rank-12 teacher	0.290	–	0.474
F-MNIST	amp.	1-epoch teacher	0.228	–	0.358

Table S16. Complete exact-gradient training results: test accuracy, mean ± s.d., followed by the number of seeds in parentheses. E2E uses the row's wiring.

Data	Enc.	Targets	E2E	LD-chain	LD-TF	LT-codes
MNIST	int.	Native	99.39 ± 0.01 (3)	98.78 ± 0.15 (3)	14.60 ± 8.46 (3)	98.85 ± 0.12 (3)
MNIST	int.	Optimized	99.42 ± 0.05 (5)	99.37 ± 0.03 (5)	97.24 ± 0.59 (5)	98.78 ± 0.09 (3)
MNIST	int.	ONN teacher	99.39 ± 0.01 (3)	99.40 ± 0.02 (3)	99.40 ± 0.02 (3)	–
MNIST	amp.	Native	99.31 ± 0.04 (3)	96.71 ± 0.16 (3)	14.73 ± 5.08 (3)	97.05 ± 0.04 (3)
MNIST	amp.	Output optimized	99.34 ± 0.01 (3)	98.58 ± 0.08 (3)	79.34 ± 4.17 (3)	–
MNIST	amp.	Grouping optimized	99.39 ± 0.02 (3)	97.37 ± 0.14 (3)	10.23 ± 1.15 (3)	–
MNIST	amp.	Both optimized	99.33 ± 0.12 (3)	98.80 ± 0.05 (3)	86.85 ± 1.32 (3)	97.24 ± 0.04 (3)
MNIST	amp.	ONN teacher	99.31 ± 0.04 (3)	99.29 ± 0.03 (3)	99.31 ± 0.06 (3)	–
F-MNIST	int.	Native	92.86 ± 0.08 (3)	89.58 ± 0.10 (3)	8.00 ± 3.31 (3)	89.77 ± 0.33 (3)
F-MNIST	int.	Optimized	92.87 ± 0.06 (5)	92.34 ± 0.16 (5)	66.33 ± 8.54 (5)	89.66 ± 0.92 (3)
F-MNIST	int.	ONN teacher	92.86 ± 0.08 (3)	92.84 ± 0.06 (3)	92.80 ± 0.11 (3)	–
F-MNIST	amp.	Native	92.61 ± 0.17 (3)	77.40 ± 0.57 (3)	8.90 ± 1.95 (3)	79.02 ± 0.43 (3)
F-MNIST	amp.	Output optimized	92.68 ± 0.15 (3)	85.06 ± 0.33 (3)	37.74 ± 5.48 (3)	–

Data	Enc.	Targets	E2E	LD-chain	LD-TF	LT-codes
F-MNIST	amp.	Grouping optimized	92.70 ± 0.05 (3)	79.18 ± 0.25 (3)	8.37 ± 2.20 (3)	–
F-MNIST	amp.	Both optimized	92.53 ± 0.19 (3)	86.43 ± 0.35 (3)	41.88 ± 3.12 (3)	79.57 ± 0.35 (3)
F-MNIST	amp.	ONN teacher	92.61 ± 0.17 (3)	92.57 ± 0.18 (3)	92.61 ± 0.21 (3)	–
F-MNIST	amp.	Sigmoid teacher	92.29 ± 0.17 (3)	89.96 ± 0.12 (3)	17.29 ± 6.24 (3)	–
F-MNIST	amp.	Dense front end	92.73 ± 0.21 (3)	81.75 ± 0.55 (3)	11.91 ± 3.76 (3)	–
F-MNIST	amp.	Rank-12 teacher	92.61 ± 0.17 (3)	72.23 ± 0.33 (2)	–	–
F-MNIST	amp.	1-epoch teacher	–	81.78 ± 0.25 (2)	–	–

Table S17. Complete target-family SPSA results: test accuracy, mean ± s.d., with the number of seeds in parentheses.

Data	Enc.	Targets	E2E-SPSA	LD-chain-SPSA
MNIST	int.	Native	99.05 ± 0.05 (3)	98.08 ± 0.19 (3)
MNIST	int.	Optimized	99.04 ± 0.09 (5)	98.96 ± 0.14 (5)
MNIST	amp.	Native	98.72 ± 0.05 (3)	92.12 ± 0.52 (3)
F-MNIST	int.	Native	91.69 ± 0.09 (3)	88.58 ± 0.17 (3)
F-MNIST	int.	Optimized	91.78 ± 0.13 (5)	91.10 ± 0.42 (5)
F-MNIST	int.	ONN teacher	91.69 ± 0.09 (3)	92.60 ± 0.05 (3)
F-MNIST	amp.	Native	90.31 ± 0.21 (3)	69.34 ± 0.40 (3)
F-MNIST	amp.	Both optimized	90.85 ± 0.25 (3)	80.22 ± 0.51 (3)
F-MNIST	amp.	ONN teacher	90.31 ± 0.21 (3)	91.46 ± 0.21 (3)

Amplitude encoding (internally pre-specified). With the native assignment the deficit of LD-chain is larger (*2.60* pp and *15.22* pp) and LD-TF collapses (*14.73* % and *8.90* %). Recalibrating the electronic readout on the student chain before inference does not rescue LD-TF (scalar gain: *8.91*%; per-port affine: *20.37*% on Fashion-MNIST), so the collapse is not a matter of scale. Optimizing only the output assignment lowers the fitted residual to *0.15* and the Fashion-MNIST deficit to *7.62* pp (LD-TF *37.74*%); optimizing only the input grouping leaves the residual at *0.54* and the deficit at *13.52* pp (LD-TF *8.37*%); optimizing both gives *0.12*, *6.09* pp and *41.88*% (improvement of LD-chain over the native assignment [+ *7.97*, + *10.10*] pp). On MNIST the deficits are *0.76* pp (output only), *2.02* pp (grouping only) and *0.53* pp (both). In this amplitude-encoded ablation (both data sets), the output assignment

accounted for most of the improvement; the split was not run with intensity encoding. With the ONN teacher LD-chain and LD-TF match E2E. Table S18 is the amplitude-encoding counterpart of main-text Table 3.

Other electronic teachers (amplitude encoding, Fashion-MNIST) mostly follow the same ordering (residuals are normalized by the uncentered target energy, which makes them smaller for targets with a large constant part, such as the sigmoid teacher, so residuals of different teachers are only roughly comparable): a sigmoid teacher with dense targets (fitted residual *0.38*) leaves *2.33* pp, a dense front end (*0.50*) *10.97* pp. There are exceptions: the rank-12 head teacher has a smaller residual (*0.47*) than the converged teacher (*0.51*) but a larger deficit (*20.38* pp, two seeds), and with the optimized assignment the residual falls to *0.12* while LD-chain still trails by *6.09* pp and LD-TF reaches only *41.88*%.

Table S18. Amplitude-encoding counterpart of main-text Table 3. The assignment-free bound uses the checked residual-space dual; the fitted residual uses the clipped readout.

Layer	Assignment/grouping	LB^{lin}	LB^{clamp}	LB^{free}	G_{TF}
MNIST L1	Native	0.371	0.335	0.151	0.593
MNIST L1	Optimized	0.097	0.059	0.112	0.145
MNIST L2	Native	0.531	0.270	0.099	0.538
MNIST L2	Optimized	0.075	0.056	0.065	0.078
F-MNIST L1	Native	0.349	0.283	0.164	0.565
F-MNIST L1	Optimized	0.105	0.062	0.131	0.154
F-MNIST L2	Native	0.280	0.187	0.102	0.459
F-MNIST L2	Optimized	0.057	0.030	0.074	0.095
F-MNIST L3	Native	0.316	0.172	0.093	0.515
F-MNIST L3	Optimized	0.054	0.026	0.066	0.097

Figure S1 shows the internally pre-specified amplitude-encoding matrix and the secondary CIFAR-10 and FSDD protocols; Table S19 lists the accuracies. These controls compare target choices while keeping the native assignment.

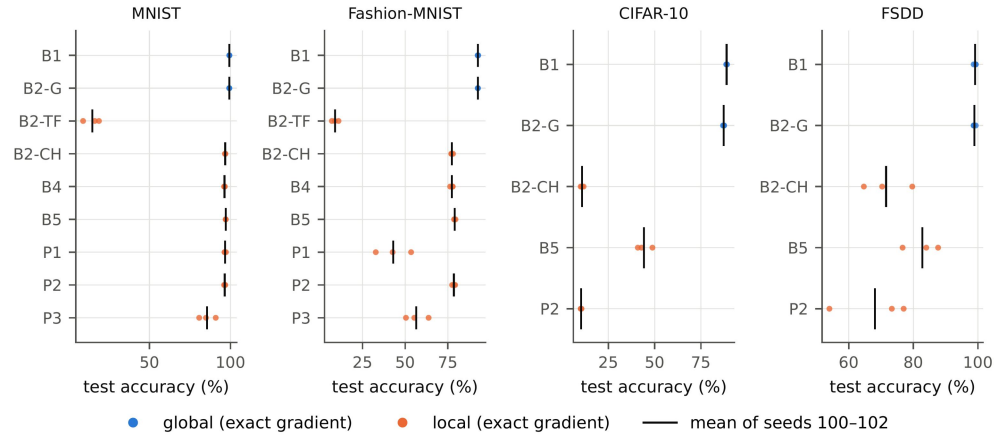


Fig. S1. Exact-gradient test accuracy with amplitude encoding and native assignment. Dots are seeds 100–102; vertical bars are their means. MNIST and Fashion-MNIST show nine methods;

the secondary CIFAR-10 and FSDD protocols include five methods. Method codes are defined in Sec. S1.

Table S19. Test accuracy (%, mean ± s.d., seeds 100–102) of the internally pre-specified matrix with exact gradients (amplitude encoding, native assignment; original codes in brackets). CIFAR-10 and FSDD were run for five methods only (–: not run).

Method	Gradient	MNIST	Fashion-MNIST	CIFAR-10	FSDD
E2E [B1]	global	99.31 ± 0.04	92.61 ± 0.17	88.86 ± 0.21	99.11 ± 0.38
LD-G [B2-G]	global	99.27 ± 0.06	92.59 ± 0.05	87.29 ± 0.34	98.89 ± 0.38
LD-TF [B2-TF]	local	14.73 ± 5.08	8.90 ± 1.95	–	–
LD-chain [B2-CH]	local	96.71 ± 0.16	77.40 ± 0.57	10.77 ± 0.80	71.56 ± 7.57
LD-chain→CE [B4]	local	96.28 ± 0.23	77.32 ± 0.86	–	–
LT-codes [B5]	local	97.05 ± 0.04	79.02 ± 0.43	44.20 ± 4.10	82.78 ± 5.60
self-dist. [P1]	local	96.63 ± 0.49	42.95 ± 10.35	–	–
self-dist. [P2]	local	96.44 ± 0.38	78.51 ± 0.91	10.22 ± 0.23	68.11 ± 12.36
self-dist.+noise [P3]	local	85.49 ± 5.13	56.51 ± 6.71	–	–

S6. Zeroth-order training and resource accounting

Figure S2 shows SPSA trajectories for the original amplitude-encoding setting, so that electronic-teacher, random-code and EMA-target variants can be compared over the training budget alongside the endpoint results and trajectories for intensity encoding in main-text Table 6 and Fig. 9; Table S20 gives the amplitude-encoding endpoints.

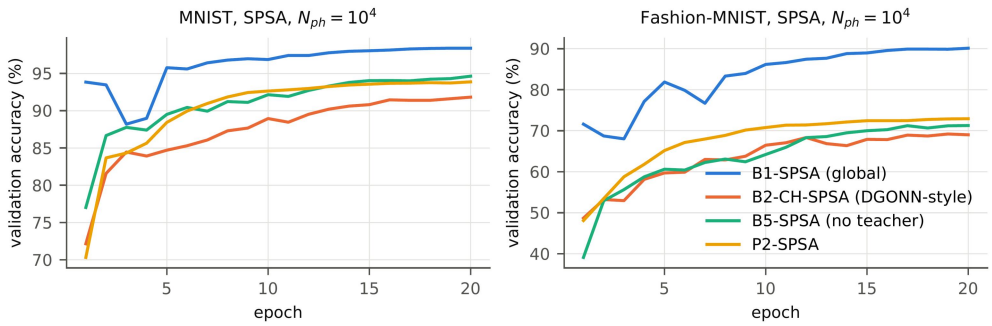


Fig. S2. Validation accuracy during SPSA training with amplitude encoding and native assignment on MNIST and Fashion-MNIST. Curves are means of seeds 100–102. B1, B2-CH, B5 and P2 denote E2E, LD-chain, LT-codes and same-depth EMA self-distillation, respectively. Measurement noise uses the Gaussian approximation with $N_{ph} = 10^4$.

Table S20. Zeroth-order (SPSA) training with the electronic teacher, amplitude encoding and native assignment (*51* (MNIST) and *68* (Fashion-MNIST) layer evaluations per step, $N_{ph} = 10^4$ unless stated). Test accuracy, %.

Method	Updates	Teacher	MNIST	Fashion-MNIST
E2E-SPSA [B1]	global	no	98.72 ± 0.05	90.31 ± 0.21
LD-chain-SPSA [B2-CH]	local	yes	92.12 ± 0.52	69.34 ± 0.40

Method	Updates	Teacher	MNIST	Fashion-MNIST
LT-codes-SPSA [B5]	local	no	95.05 ± 0.27	71.18 ± 0.37
SD-EMA-SPSA [P2]	local	annealed	94.40 ± 0.39	72.92 ± 2.98
E2E-SPSA [B1] (no shot noise)	global	no	98.60 ± 0.07	90.49 ± 0.07
LD-chain-SPSA [B2-CH] (no shot noise)	local	yes	92.58 ± 1.06	71.63 ± 0.80

The original coordinate-difference method [1] differs from SPSA. Our coordinate comparison at 6×10^5 layer measurements favors global differences (*67.87*% versus *55.35*%, amplitude encoding), but uses simultaneous block perturbations and covers two seeds. Figure S3 displays the coordinate-difference and SPSA trajectories on a common layer-measurement axis; this accounting separates the update rule from the number of optical measurements used to obtain each update. Detected-photon totals need not be equal even at equal nominal exposure. With the optimized intensity-encoding assignment, equal-grid tuning leaves a Fashion-MNIST deficit of *0.30* pp.

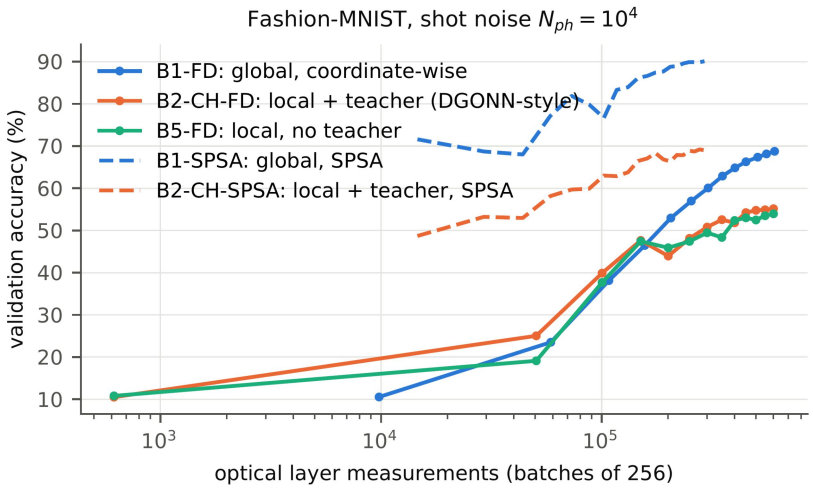


Fig. S3. Validation accuracy versus optical layer measurements on Fashion-MNIST with amplitude encoding and native assignment. Solid curves show coordinate-wise forward differences averaged over two seeds; dashed curves show SPSA averaged over three seeds. Measurements use batches of 256 and Gaussian-approximated shot noise with $N_{ph} = 10^4$. The coordinate comparison uses simultaneous block perturbations.

Table S22 gives the tuned comparison (amplitude encoding, native assignment); for intensity encoding with the optimized assignment the selected settings are 0.01×40 epochs (E2E, *92.79*%) and 0.03×40 epochs (LD-chain, *92.49*%). SPSA learning rates were selected per method on the validation set from {*0.003*,*0.01*,*0.03*,*0.1*} (B1-SPSA extended to 0.3 after its optimum lay at 0.1); the optima of all local methods lie inside the grid. SPSA uses *51* (MNIST, three optical layers) and *68* (Fashion-MNIST, four optical layers) layer measurements per update for both global and local estimators, so within each data set measurement count, update count and nominal exposure (the same N_{ph} per unit detected intensity) are matched. The number of detected photons is not matched, because shot-noise variance scales with the detected intensity and the detected fraction of the light differs between methods (after training, mean $\eta = 0.60$ for E2E-SPSA and *0.56* for LD-chain-SPSA, intensity encoding with optimized assignment). The following counts refer to the Fashion-MNIST architecture only (*272* subnetworks): our coordinate-wise scheme uses *616* (local) and *9796* (global) layer measurements per update and was compared at equal total

measurements (different update counts); the per-coordinate scheme of Ref. [1] needs *4081* network evaluations per NDNN update and *4084* layer evaluations per DGONN update with a layer-parallel depth of *1921*; the resulting time factor *2.1* assumes simultaneous measurement of all layers and measurement-dominated time, and it applies to coordinate-wise differences only. If a network pass cannot be pipelined and costs one time unit per layer, the same coordinate-wise factor becomes $4081 \times 4/1921 \approx 8.5$. For SPSA the factor is 1 under the convention of Ref. [1], so global and local SPSA both cost *17* time units per update; without pipelining it would be at most *2.7* (MNIST) and *3.4* (Fashion-MNIST). Table S21 collects the zeroth-order comparisons by number of updates; the 154-epoch runs are an update-matched sensitivity analysis, not a test of convergence. As a sensitivity analysis for that case, LD-chain-SPSA was trained with 2.1 and 4 times the steps in the reference setting (intensity encoding, optimized assignment, five seeds): paired differences to E2E-SPSA with the original steps are $[+0.04, +0.19]$ (MNIST) and $[-0.49, +0.26]$ pp (Fashion-MNIST) at 2.1 times and $[+0.01, +0.27]$ and $[-0.25, +0.41]$ pp at 4 times. These runs were planned with the coordinate-wise factor 2.1, whose earlier use as a time factor for SPSA was an error (Sec. S10); we draw no conclusion about Ref. [14] from them.

Table S21. Update-matched sensitivity analysis of zeroth-order training with the electronic teacher, intensity encoding (test accuracy, %, MNIST / Fashion-MNIST; paired two-sided 95% intervals of LD-chain-SPSA − E2E-SPSA in pp).

Protocol	updates	E2E-SPSA	LD-chain-SPSA	difference
7000/3000, native, 20 epochs, 3 seeds	*560*	*96.82 / 86.02*	*93.04 / 78.97*	$[-4.99, -2.57]$ / $[-12.53, -1.58]$
7000/3000, native, 154 epochs, 3 seeds	*4312*	*97.72 / 87.69*	*96.64 / 84.98*	$[-1.45, -0.70]$ / $[-3.67, -1.75]$
full sets, optimized, 20 epochs, 5 seeds	*4300*	*99.04 / 91.78*	*98.96 / 91.10*	$[-0.19, +0.03]$ / $[-1.29, -0.08]$

Table S22. Methods tuned on the same grid (learning rate × epochs, seed 100, validation set); test accuracy over seeds 100–104 (amplitude encoding, native assignment).

Data	Method	selected (lr, epochs)	accuracy (5 seeds)	gap to tuned E2E
F-MNIST	E2E	(0.01, 40)	92.45 ± 0.24	0.00
F-MNIST	LD-G	(0.01, 40)	92.60 ± 0.06	-0.15
F-MNIST	LD-chain	(0.01, 40)	78.09 ± 0.56	14.36
F-MNIST	LT-codes	(0.03, 40)	80.07 ± 0.76	12.38
MNIST	E2E	(0.003, 20)	99.25 ± 0.05	0.00
MNIST	LD-G	(0.003, 40)	99.17 ± 0.02	0.08
MNIST	LD-chain	(0.03, 40)	97.03 ± 0.09	2.22
MNIST	LT-codes	(0.03, 40)	97.44 ± 0.18	1.81

S7. Fine-tuning on simulated chips

We transfer trained E2E parameters to simulated chips with static phase errors ($\sigma_\phi = 0.05\pi$) and coupler imbalance ($\sigma = 0.03$), then compare global SPSA with LD-chain-SPSA toward the ideal digital model's layer outputs. Both start from the same transferred parameters and use equal measurement counts. Three networks are crossed with three chip realizations; a

separate chip selects learning rates. The conservative crossed-design intervals remain positive only for Fashion-MNIST after three epochs: $[+0.03, +0.46]$ pp with intensity encoding and $[+0.29, +0.79]$ pp with amplitude encoding (Table S23). Main-text Fig. 10 shows the twelve-epoch validation trajectories for both encodings. This exploratory simulation supports target-based adaptation within those budgets; it neither establishes a benefit at convergence nor constitutes a physical-chip experiment.

For each configuration the nine differences Δ_{ij} (network i, chip j) are analysed with the balanced two-way random-effects model $\Delta_{ij} = \mu + u_i + v_j + \varepsilon_{ij}$ (one observation per cell, interaction absorbed in ε). With mean squares $\mathrm{MS_{net}}$ and $\mathrm{MS_{chip}}$ (two degrees of freedom each) and $\mathrm{MS_{err}}$ (four), the variance components are $\sigma_u^2 = (\mathrm{MS_{net}} - \mathrm{MS_{err}})/3$, $\sigma_v^2 = (\mathrm{MS_{chip}} - \mathrm{MS_{err}})/3$ and $\sigma_\varepsilon^2 = \mathrm{MS_{err}}$, with negative estimates set to zero, and $\mathrm{Var}(\hat{\mu}) = \sigma_u^2/3 + \sigma_v^2/3 + \sigma_\varepsilon^2/9$. This equals $(\mathrm{MS_{net}} + \mathrm{MS_{chip}} - \mathrm{MS_{err}})/9$ when both components are positive and $\mathrm{MS_{chip}}/9$ or $\mathrm{MS_{net}}/9$ when one is truncated; the degrees of freedom follow Satterthwaite after the coefficients of each mean square are combined. The rule was fixed before the computation. To see how much the conclusions depend on the variance approximation, we also use the estimate $(\mathrm{MS_{net}} + \mathrm{MS_{chip}})/9$, whose expectation exceeds $\mathrm{Var}(\hat{\mu})$ by $\sigma_\varepsilon^2/9$, with two degrees of freedom, the lower limit of the Satterthwaite approximation for a sum of two mean squares with two degrees of freedom each. These conservative intervals stay above zero only for Fashion-MNIST at three epochs, with both encodings (Table S23); the other positive crossed-model intervals depend on the approximation. An earlier version of this check paired the conservative variance with Satterthwaite degrees of freedom (2–4), which made it narrower than the main interval in seven of the eight cases; its statement that all eight signs agree was an error and is withdrawn. Applying the Satterthwaite formula to the untruncated combination (negative coefficient on $\mathrm{MS_{err}}$), or treating the two occurrences of $\mathrm{MS_{err}}$ as independent terms, gives about one degree of freedom or fewer and very wide intervals exactly in the cells with a negative component estimate; we regard this as instability of the approximation, not as additional variation in the data. A two-way cluster bootstrap is not used because each factor has only three levels.

Table S23. Device-error study on simulated chips: mean difference LD-chain-SPSA − E2E-SPSA (pp) with crossed random-effects intervals (truncated components, Satterthwaite df) and conservative intervals (variance $(\mathrm{MS_{net}} + \mathrm{MS_{chip}})/9$, two degrees of freedom).

Encoding	epochs	Data	Δ	95% CI (crossed)	df	95% CI (conservative, df = 2)
$a = x$	3	MNIST	*+0.11*	$[+0.02, +0.19]$	*3.2*	$[-0.01, +0.23]$
		F-MNIST	*+0.54*	$[+0.37, +0.71]$	*4.0*	$[+0.29, +0.79]$
$a = x$	12	MNIST	*+0.10*	$[+0.02, +0.18]$	*2.0*	$[-0.00, +0.19]$
		F-MNIST	*+0.38*	$[+0.06, +0.71]$	*2.2*	$[-0.01, +0.78]$
intensity	3	MNIST	*+0.03*	$[-0.02, +0.09]$	*4.0*	$[-0.02, +0.09]$
		F-MNIST	*+0.25*	$[+0.04, +0.46]$	*2.0*	$[+0.03, +0.46]$
intensity	12	MNIST	*+0.04*	$[-0.02, +0.10]$	*4.0*	$[-0.03, +0.11]$
		F-MNIST	*+0.24*	$[-0.14, +0.63]$	*2.0*	$[-0.14, +0.63]$

Table S24. Exploratory crossed-chip experiment: fine-tuning on simulated chips with static errors (three networks × three chips; test accuracy on the simulated chip, %, mean of the nine pairs). Transfer: parameters

of the ideal network without training; E2E-SPSA and LD-chain-SPSA (toward the layer outputs of the ideal digital model) start from the transferred parameters with the same measurement count; Δ: LD-chain-SPSA − E2E-SPSA (pp); interval: two-sided 95% interval for the mean difference under the crossed random-effects model (not adjusted for multiplicity). Raw *3 × 3* results are retained with the simulation records.

Encoding	Data	epochs	ideal	transfer	E2E-SPSA	LD-chain-SPSA	Δ	95% CI (crossed)	df
$a = x$	MNIST	3	99.31	98.85	99.10	99.21	*+0.11*	[*+0.02*, *+0.19*]	3.2
$a = x$	F-MNIST	3	92.61	88.88	91.96	92.50	*+0.54*	[*+0.37*, *+0.71*]	4.0
$a = x$	MNIST	12	99.31	98.85	99.18	99.27	*+0.10*	[*+0.02*, *+0.18*]	2.0
$a = x$	F-MNIST	12	92.61	88.88	92.16	92.54	*+0.38*	[*+0.06*, *+0.71*]	2.2
$a = \sqrt{x}$	MNIST	3	99.39	98.95	99.32	99.35	*+0.03*	[*−0.02*, *+0.09*]	4.0
$a = \sqrt{x}$	F-MNIST	3	92.86	89.32	92.53	92.78	*+0.25*	[*+0.04*, *+0.46*]	2.0
$a = \sqrt{x}$	MNIST	12	99.39	98.95	99.34	99.38	*+0.04*	[*−0.02*, *+0.10*]	4.0
$a = \sqrt{x}$	F-MNIST	12	92.86	89.32	92.58	92.82	*+0.24*	[*−0.14*, *+0.63*]	2.0

S8. Optical power and offline robustness

The secondary amplitude-encoding controls characterize the optical operating conditions associated with different local objectives. Figure S4 pairs classification accuracy with detected-to-input power and the fraction of small hidden outputs. The power-regularized control directs more input power to detected ports, while the depth profiles reveal how the local objectives distribute activity through the network; fitting sparse targets routes light into filtered ports. Figure S5 evaluates accuracy retention under 19 non-ideality conditions without adaptation to the evaluation chip. Here κ is each network's non-ideal accuracy divided by its ideal accuracy, so the retention ratio and absolute accuracy describe different aspects of transfer performance. Annealing the teacher weight toward EMA self-distillation targets (P1–P3, Sec. S1) does not substitute for the teacher in hidden layers under local updates (their errors match those of the truncated teacher within 10^{-3}, H1b). Physics-in-the-loop training (P3) retains a larger fraction of its ideal accuracy than LD-chain (H3), but κ measures offline transfer to chips with static errors, not on-chip training, and E2E combines the highest retention with the highest absolute accuracy (Table S25).

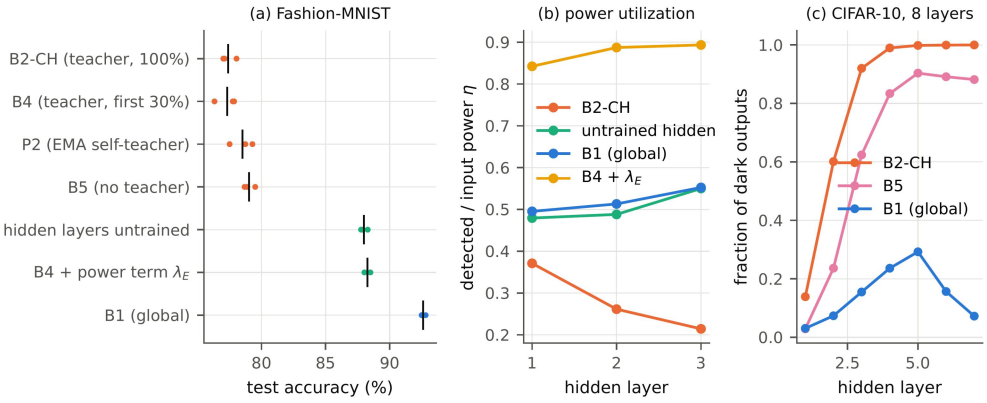


Fig. S4. Exploratory optical-power diagnostics with amplitude encoding and native assignment. (a) Fashion-MNIST accuracy for local-update variants, untrained hidden layers and E2E

training; dots are seeds 100–102 and vertical bars are their means. (b) Detected-to-input power per hidden layer on Fashion-MNIST. (c) Fraction of hidden outputs below 10^{-3} versus depth on CIFAR-10. Panels (b,c) use seed 100 and noiseless test-set inference. The additional power term is $\lambda_E(1 - \text{mean}\,\eta)$, with $\lambda_E = 0.05$. The small-output threshold differs from the exactly zero input blocks used in the residual certificates.

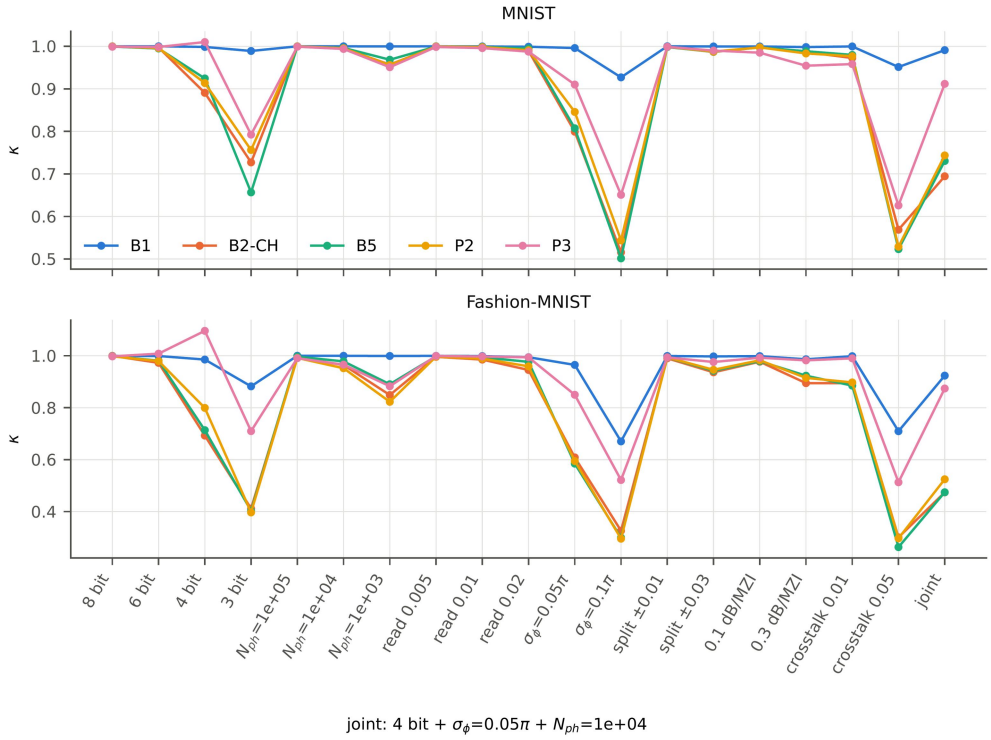


Fig. S5. Offline accuracy retention κ under 19 non-ideality conditions, with amplitude encoding and native assignment. Values are means over seeds 100–102 of each model's non-ideal-to-ideal accuracy ratio. Each non-ideal evaluation averages three chip realizations, with two measurement-noise repetitions when applicable. Parameters are not fine-tuned to the evaluation chip. B1, B2-CH, B5, P2 and P3 denote the methods defined in Sec. S1.

Table S25. Joint non-ideality (4-bit phases, $\sigma_\phi = 0.05\pi$, $N_{ph} = 10^4$): ideal accuracy (%), retention κ and absolute accuracy (%), amplitude encoding (original codes in brackets).

	MNIST			Fashion-MNIST		
Method	ideal	κ	ideal× κ	ideal	κ	ideal× κ
E2E [B1]	99.31	0.991	98.42	92.61	0.923	85.52
LD-G [B2-G]	99.27	0.984	97.68	92.59	0.898	83.18
LD-chain [B2-CH]	96.71	0.694	67.16	77.40	0.474	36.68
LT-codes [B5]	97.05	0.730	70.88	79.02	0.474	37.49
self-dist. [P2]	96.44	0.743	71.70	78.51	0.524	41.17
self-dist.+noise [P3]	85.49	0.912	77.94	56.51	0.874	49.38

S9. DGONN reconstruction and comparison

Table S26 compares our model with Ref. [14] and its supplement [1]. We use the reported layer widths, 4×4 Reck blocks with 15 phases each (3 output phases are inert under intensity detection, so 12 phases per block matter), port filtering for reduction and a three-module summed output layer; these reproduce the reported phase counts (3600 and 4080), which were used to calibrate the reconstruction and are therefore not independent evidence. Two choices are not determined by the main text of Ref. [14]. First, it encodes normalized features as field amplitudes, $\cos(\theta/2)$ in its Eq. (6), whereas Eq. (S57) of the supplement loads $\sqrt{|x|}$, i.e. intensity encoding. Intensity encoding is therefore the reference setting; the amplitude-encoded network, with which the internally pre-specified experiments were run before the supplement was available to us, is reported as a second architecture. Second, Eq. (S57) has no electronic gain or clipping, while the main text mentions an electronic activation and normalization to $[0,1]$. Our readout $\mathrm{clamp}(sI, 0, 1)$ is not equivalent to the unclipped model even at $s = 1$, because they differ wherever $sI > 1$ (in the LD-chain networks this concerns at most 0.74% of hidden outputs with amplitude encoding and 0.17% and 0.44% with intensity encoding and the native and optimized assignment). For the unclipped model of Eq. (S57), $\mathrm{LB}^{\mathrm{lin}}$ is a valid, conservative certificate: it covers every scalar gain, including $s = 1$. We do not fix $s = 1$, because the scale of the targets is set by our normalization (99.5th percentile), so a certificate at fixed s would depend on that arbitrary choice.

Table S26. Reconstruction compared with Ref. [14] (main text, M) and its Supplement 1 [1] (S). Accuracies are test accuracies in % (MNIST / Fashion-MNIST); ours under the 7000/3000 protocol of Ref. [14], mean of three seeds, with intensity encoding (reference) and, in brackets, amplitude encoding. Teacher accuracies do not depend on the encoding.

Item	Ref. [14]	This work
Blocks, phases	4×4 Reck, 15 phases per block (M)	same; 3 output phases inert
Layer widths	512-256-64-10, 512-256-128-64-10 (M)	same
Reduction, output layer	port filtering; "three 64-dim modules" (M); Eq. (S58) ambiguous	port filtering; 3 modules, ports summed
Loading	amplitude $\cos(\theta/2)$ (M, Eq. 6); $\sqrt{\lvert x\rvert}$ (S, Eq. S57)	intensity encoding (reference); $a = x$ (internally pre-specified)
Electronic stage	"activation", normalization to $[0,1]$ (M); none in Eq. (S57)	gain s and clamp to $[0,1]$
Front end	DLA-type CNN (S6)	small CNN
Layer loss, update	MSE per layer; FD per coordinate, momentum (S5, S8)	MSE per layer, τ^2KL+CE at output; exact, SPSA or coordinate FD
Unit-to-port assignment	not discussed	native and optimized
Data protocol	7000/3000 random split (M)	full sets (main), 7000/3000 (this table)
Teacher accuracy	99.4 / 94.73 (M, Table 1)	*98.33* / *88.70* (7000/3000); *99.53* / *93.44* (full sets)
DGONN accuracy	98.87 / 94.73 (M, Table 1)	native assignment: LD-chain *96.57* / *84.91* (*93.61* / *75.10*); LD-G *98.22* / *88.39* (*98.06* / *87.60*); LD-chain-SPSA *93.04* / *78.97* (*82.11* / *56.90*)
NDNN accuracy	90.67 / 91.33 (M, Table 1)	E2E-SPSA *96.82* / *86.02* (*96.91* / *85.20*)

Item	Ref. [14]	This work
SPSA budget	–	20 epochs = *560* updates (full sets: *4300*); update-matched sensitivity analysis, 154 epochs (*4312* updates), intensity encoding, native assignment: E2E-SPSA *97.72* / *87.69*, LD-chain-SPSA *96.64* / *84.98* (Table S21)
Optimized assignment	–	intensity encoding: E2E *98.22* / *88.68*, LD-chain *97.67* / *86.28*, E2E-SPSA *96.97* / *85.37*, LD-chain-SPSA *93.62* / *79.02*

With the zeroth-order estimators and budgets we tested, our reconstruction does not reproduce the advantage of layer-wise distillation over non-distilled forward-difference training reported in Ref. [14]. At equal measurement count, which for SPSA also means equal time under the convention of Ref. [1] (Sec. S6), LD-chain-SPSA with the electronic teacher does not exceed E2E-SPSA in any of our settings. In the 7000/3000 protocol of Ref. [14] with intensity encoding, 20 epochs correspond to only *560* SPSA updates (*4300* with the full training sets); after 20 epochs LD-chain-SPSA trails E2E-SPSA (paired differences $[-4.99, -2.57]$ and $[-12.53, -1.58]$ pp with the native and $[-4.58, -2.10]$ and $[-8.26, -4.43]$ pp with the optimized assignment, three seeds). With the number of updates matched to the full-data runs (154 epochs, *4312* updates) the gap shrinks but remains ($[-1.45, -0.70]$ and $[-3.67, -1.75]$ pp, native assignment), so part of the 20-epoch deficit reflects the small budget. Our coordinate-wise comparison approximates, but does not exactly reproduce, the per-coordinate scheme with momentum of Ref. [1]. This is a failure to reproduce, not a demonstration that the original result is wrong, because our reconstruction differs in the front end, the output layer and possibly the training details (Table S26). Under the 7000/3000 protocol with intensity encoding and the native assignment, per data set and method: our teacher is -1.07 and -6.03 points from the reported teacher accuracy, our LD-chain -2.30 and -9.82 and our LD-G -0.65 and -6.34 points from the reported DGONN accuracy, and our E2E-SPSA $+6.15$ and -5.31 points from the reported NDNN accuracy (amplitude encoding, LD-chain and E2E-SPSA: -5.26, -19.63, $+6.24$ and -6.13). The weaker Fashion-MNIST teacher can account for part of the LD-chain difference but not for all of it, and testing other explanations would require controlled experiments with the original front end. In terms of the teacher-recovery ratio of Table 1 of Ref. [14] (99–100%), with intensity encoding our LD-G recovers *99.9*% and *99.6*% of its teacher and LD-chain *98.2*% and *95.7*% (*99.3*% and *97.3*% with the optimized assignment). Training with the global gradient is therefore one possible explanation of part of the difference; Ref. [14] describes both local updates and "end-to-end optimization", and which regime produced its numerical results cannot be decided from the publication. The unit-to-port assignment, not discussed in Ref. [14], strongly affects layer-wise training in our simulations. The on-chip experiment of Ref. [14] is outside our scope.

S10. Internally pre-specified tests, gates and interpretation rules

Gates. K1 original criterion: student accuracy on planted-teacher labels $\geq 99.7\%$; amended: accuracy on samples whose top-2 logit margin exceeds one class-standard-deviation $\geq 99.7\%$ (attained *99.91*% and *99.94*%) plus hidden-layer residual $< 5 \times 10^{-3}$. K2 original: shuffled-teacher $G_1 \geq 3 \times$ genuine; amended: shuffled $G_1 >$ genuine G_1, because the genuine teacher is already above the constant-predictor residual. Both amendments were made after the failures and before the main experiments. K5 and H1 were rewritten after the validation pilot and before freezing. The original amplitude-encoding, native-assignment comparisons were internally pre-specified (frozen protocol with a SHA-256 hash recorded with the simulation records) before their test evaluation, with one-sided paired tests and Holm correction over ten

tests. Later analyses, including the intensity-encoding comparisons, are post hoc; their interpretation rules were recorded before execution.

Tests. H1a compares LT-codes minus LD-chain with a -1 pp margin; H3 compares retention κ of P3 minus LD-chain with zero; H4 compares power utilization η of P2 minus LD-chain with a -0.02 margin; H5a compares LD-chain-SPSA minus E2E-SPSA with zero; H5b compares LT-codes-SPSA minus LD-chain-SPSA with a -1 pp margin. Table S27 follows the frozen seed rule: H1a uses seeds 100–106, whereas H3, H4, H5a and H5b use 100–102. Table S28 adds seeds 103–106 to the latter tests post hoc; H1a is unchanged. Both tables report effect sizes and raw and Holm-adjusted p-values. Confidence intervals are two-sided, p-values one-sided as internally pre-specified. "Not supported" denotes a one-sided test that does not pass; it does not establish the opposite direction, for which we only report post hoc two-sided intervals.

Table S27. Internally pre-specified tests (differences in pp; H3: $\kappa \times 100$; H4: $\eta \times 100$).

Test	Data	n	Δ [95% CI]	d_z	p	p_{Holm}
H1a	MNIST	7	$+0.48$ [$+0.22, +0.73$]	1.7	< 0.001	< 0.001
H1a	F-MNIST	7	$+0.88$ [$-0.63, +2.38$]	0.5	0.011	0.047
H3	MNIST	3	$+21.73$ [$+8.69, +34.76$]	4.1	0.009	0.047
H3	F-MNIST	3	$+40.00$ [$+9.15, +70.84$]	3.2	0.015	0.047
H4	MNIST	3	$+1.27$ [$+0.49, +2.05$]	4.0	0.002	0.011
H4	F-MNIST	3	$+2.60$ [$+2.12, +3.07$]	13.6	< 0.001	0.003
H5a	MNIST	3	-6.60 [$-7.89, -5.31$]	-12.7	0.999	> 0.999
H5a	F-MNIST	3	-20.96 [$-21.77, -20.16$]	-64.6	> 0.999	> 0.999
H5b	MNIST	3	$+2.93$ [$+2.19, +3.66$]	9.9	< 0.001	0.008
H5b	F-MNIST	3	$+1.84$ [$+1.15, +2.52$]	6.6	0.002	0.011

Table S28. Post hoc extension of the tests to seven seeds. Hypothesis definitions and units are the same as in the preceding table; p values are one-sided and intervals two-sided.

Test	Data	n	Δ [95% CI]	d_z	p	p_{Holm}
H1a	MNIST	7	$+0.48$ [$+0.22, +0.73$]	1.7	< 0.001	< 0.001
H1a	F-MNIST	7	$+0.88$ [$-0.63, +2.38$]	0.5	0.011	0.034
H3	MNIST	7	$+20.81$ [$+16.41, +25.21$]	4.4	< 0.001	< 0.001
H3	F-MNIST	7	$+37.35$ [$+29.98, +44.73$]	4.7	< 0.001	< 0.001
H4	MNIST	7	$+1.33$ [$+1.11, +1.54$]	5.6	< 0.001	< 0.001
H4	F-MNIST	7	$+3.01$ [$+2.43, +3.59$]	4.8	< 0.001	< 0.001
H5a	MNIST	7	-6.21 [$-6.84, -5.59$]	-9.2	> 0.999	> 0.999
H5a	F-MNIST	7	-21.77 [$-22.66, -20.88$]	-22.6	> 0.999	> 0.999
H5b	MNIST	7	$+1.65$ [$+0.27, +3.03$]	1.1	0.002	0.007
H5b	F-MNIST	7	$+2.23$ [$+1.11, +3.35$]	1.8	< 0.001	0.001

Interpretation rules. Table S29 lists every interpretation rule written before the corresponding analysis, the result, the conclusion it permits and deviations. An internally pre-specified rule constrains the analysis; it does not by itself establish the scientific claim.

Table S29. Audit of internally pre-specified interpretation rules.

Analysis	Rule	Result	Permitted conclusion	Deviation
Clamp certificate	all layers certified; if $LB^{clamp} < 0.5\,LB^{lin}$ quote clamp values	certified; ratio ≥ 0.51	certified for clipped readout, given assignment	tail-interval solver refined after a valid but loose first run
Assignment (amplitude)	LD-chain $+\geq 5$ pp $\Rightarrow$ assignment explains most of the deficit; LD-TF $> 20\%$ $\Rightarrow$ "collapse" only for native assignment	$[+7.97, +10.10]$ pp; LD-TF 41.88%	as stated	grouping and output assignment first changed jointly; separated afterwards
Paper protocol (7000/3000)	LD-G $-$ LD-chain < 2 pp $\Rightarrow$ name the global-gradient regime as leading candidate	$a = x$: $4.4/12.5$ pp; intensity: $1.7/3.5$ pp	triggered for MNIST with intensity encoding; stated only as one possible explanation	rule written for $a = x$; its "leading candidate" wording was withdrawn after review, because a small LD-G–LD-chain difference limits what the training regime can explain
Paper protocol, SPSA (7000/3000, intensity)	LD-chain-SPSA $-$ E2E-SPSA CI > 0 $\Rightarrow$ "reproduced"	native: $[-4.99, -2.57]$, $[-12.53, -1.58]$; optimized: $[-4.58, -2.10]$, $[-8.26, -4.43]$	not reproduced	–
Encoding	"robust to encoding" only if all $LB^{lin} > 0.1$, LD-chain gap > 5 pp, LD-TF $\leq 20\%$	gaps $0.62/3.27$ pp	encoding dependence must be stated; "robust" not permitted	an earlier abstract claimed both encodings; withdrawn
Positive control	LD-TF $\geq$ E2E-5 pp and LD-chain $\geq$ E2E-3 pp	met	reachable task-relevant targets let layer-wise training succeed	causal wording of the rule not used
Gain rescaling	$> 50\%$ $\Rightarrow$ scale mismatch	$8.91/20.37\%$	collapse not explained by readout scale	–
Hierarchy	energy step < 0.05 $\Rightarrow$ not a main constraint	0.024	as stated	regressor estimator revised; percentage attributions withdrawn
Tuning (amplitude)	tuned gap < 5 pp $\Rightarrow$ report tuned gaps; $4\times$ steps $\geq$ E2E-SPSA-1 $\Rightarrow$ restrict H5a	MNIST 2.22 pp; 69.96%	tuned gaps reported; H5a not restricted	–
Seven seeds	–	decisions unchanged	–	–

Analysis	Rule	Result	Permitted conclusion	Deviation
Intensity × optimized	LD-chain ≥ E2E−*1* pp ⇒ "within 1 pp"; SPSA CI > *0* / ∋ *0* / < *0*; LD-TF thresholds	[− *0.68*, − *0.38*], [− *0.14*, + *0.05*]; SPSA [− *1.29*, − *0.08*], [− *0.19*, + *0.03*]; LD-TF *66.33*/*97.24*%	within 1 pp; SPSA behind (F-MNIST) / indistinguishable (MNIST); LD-TF partial / recovered	tuning grid added as specified
Output vs grouping	descriptive only	see Table S16	descriptive	–
Time alignment (intensity, optimized)	LD-chain-SPSA (*2.1* × steps) − E2E-SPSA: CI > *0* ⇒ conditional advantage	[+ *0.04*, + *0.19*], [− *0.49*, + *0.26*]; *4* ×: [+ *0.01*, + *0.27*], [− *0.25*, + *0.41*]	none; sensitivity analysis only (Sec. S6)	rule withdrawn: the factor *2.1* holds for coordinate-wise differences, not for SPSA (factor 1 under the same convention)
Update-matched SPSA (7000/3000, intensity)	LD-chain-SPSA − E2E-SPSA with *4312* updates: CI > *0* / ∋ *0* / < *0*	[− *1.45*, − *0.70*], [− *3.67*, − *1.75*]	behind also with matched updates; gap smaller than after 20 epochs	–
Seed control (ONN teacher)	LD-TF ≥ E2E−*5* pp and LD-chain ≥ E2E−*3* pp with student seeds ≠ teacher seed	*92.73*/*92.71*% vs *92.86*%	positive control does not depend on shared initialization	–
Static chip errors	second round: CI over 9 pairs > *0* ⇒ design statement; third round: CIs over networks and over chips both > *0*; fifth round (replaces the third): crossed random-effects CI > *0* ⇒ keep, ∋ *0* ⇒ "not distinguished from zero"; statement restricted to the encoding where it holds	Table S24; $a = x$: crossed CIs > *0* (3 and 12 epochs, both data sets); intensity: > *0* only for F-MNIST at 3 epochs; conservative intervals (df = *2*) > *0* only for F-MNIST at 3 epochs, both encodings	slightly higher with $a = x$ (crossed model), with intensity encoding only F-MNIST at 3 epochs; robust to the variance approximation only for F-MNIST at 3 epochs; exploratory	third-round rule replaced because intervals over network or chip means each ignore one random factor; for intensity F-MNIST at 12 epochs the third-round wording "E2E-SPSA catches up" was not used because the mean difference did not shrink; the fifth-round conservative check used Satterthwaite df and was not conservative (corrected to df = *2*)
Same-family check	every certificate below a direct fit of its own readout family, otherwise stop	*120* pairs, all ratios ≥ *1.01*	certificates consistent with the fits of each family	repeated with the implemented-encoding certificates
Block size	fit < *0.01* at $m = 64$ ⇒ "nearly reachable"	*0.011–0.028*	certificate no longer excludes reachability; fits leave residual	one positive value whose gain cap could not be removed (F-MNIST L3, $m = 16$) had been listed as certified; now counted as zero

Analysis	Rule	Result	Permitted conclusion	Deviation
Dark blocks	offset readouts use target variance	*0.010–0.047* (given assignment); minimum over assignments ≤ 0.0003	certificate for the given assignment only	earlier claims for affine readouts and for all assignments withdrawn
Dual certificate of LB^{free} (audit)	reproduction within 10^{-6} and no dual value above the primal, otherwise stop; report the dual-free signed bound	reproduced; dual-free bound *0.030–0.160*	historical numerical consistency only	superseded by the checked residual duals (Sec. S2.3)
Checked residual certificates	independently check residual duals; maximize per block and unit before aggregation	*20* positive; *0* fallbacks; at least *99.6*% of relaxed primal	assignment-free bounds	post hoc correction and strengthening of an earlier estimate; affine PSD estimates remain diagnostic (Sec. S4)
Implemented encoding	recompute with $\sqrt{x+\epsilon}-\sqrt{\epsilon}$; all bounds $> 0 \Rightarrow$ replace the ideal-encoding values; any $\leq 0 \Rightarrow$ positivity not established there	all positive; three clipped certificates on large-gain intervals change by up to *0.0085* (large Frank–Wolfe gaps, Sec. S2.4), all others by at most 5.0×10^{-6}	intensity-encoding certificates reported for the implemented encoding	the clamp runs were restarted once to store interval data; the one layer completed before the restart agreed to five decimals
Rounding-error recheck	every determining dual vector passes an exact PSD test after rounding-error bounds; otherwise drop that candidate	*730,880* tested, none failed; shortfall $\leq 1.5 \times 10^{-10}$	assignment-free bounds verified against rounding	–
Regression baseline	clipped ONN fit below Q in every layer $\Rightarrow$ "regression does not bound clipped readouts"; optimized ratio ≤ 1.25 in every layer $\Rightarrow$ "little room for reassignment"	$G_{TF} \leq 1.9 \times 10^{-4}$ vs $Q \geq 0.011$; ratio ≤ 1.23	both statements permitted	–

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